Brand Search Intelligence Platform — Product Requirements Document (PRD)

2026-09-18 · @Someone

This PRD is the single source of truth for building the Brand Search Intelligence + AI Visibility Optimization platform (working name **Lumenca**) as a production-ready, multi-tenant web application on Google Cloud Platform. It supersedes the ADK demo and builds on the transformation plan delivered on 18 Sep 2026.

1. Document control

Item	Value
Product	Brand Search Intelligence + AI Visibility Optimization platform (working name: Lumenca; final name pending trademark clearance)
Owner	Vineet, AIxAI
Audience	Product, design, engineering, DevOps, QA, security
Status	Draft v1.0 for review
Scope	MVP (Phase 1, 90 days) through Enterprise (Phase 3, 12 months)
Related	Transformation plan (18 Sep 2026); ADK demo test guide (18 Sep 2026)

1.1 Assumptions

- The existing ADK application is treated as a reference implementation only; its generator/critic loop, `google_search` grounding, BigQuery access and eval discipline are reused as internal libraries, not as the product runtime.
- AI engines are queried through official APIs with web search (OpenAI Responses `web_search`, Anthropic web search tool, Gemini grounding, Perplexity Sonar). Google SERP, AI Overviews and AI Mode data come from a licensed SERP data provider (primary: DataForSEO; fallback: SerpApi). Consumer-UI scraping is out of scope for the core pipeline.

- AI answers are probabilistic. Every visibility metric is computed from repeated samples (default 3 in MVP, 7 for Pro/Enterprise) and reported with sample size and a 95% confidence interval.
- Both India and global (US/EU) customers are targeted from day one; the beachhead is India D2C/e-commerce brands and agencies.
- One engineering pod (2 backend, 2 frontend, 1 data/AI, 1 DevOps, 1 QA, 1 designer, 1 PM) delivers Phase 1.

1.2 Constraints

- GCP only for cloud infrastructure; region pairing `asia-south1` (Mumbai) primary and `us-central1` for US-resident tenants.
- Third-party API cost must stay under 35% of revenue at the Pro tier; cost per monitored prompt is a first-class metric.
- No feature may promise rankings, AI recommendations, Knowledge Panels or citations. Proprietary scores are labelled as diagnostics.
- No fake searches, engagement, reviews or manipulation features, ever.
- India DPDP Act and GDPR compliance required before Phase 2 general availability.

1.3 Dependencies

Dependency	Type	Risk if unavailable
OpenAI, Anthropic, Google, Perplexity APIs	External	Engine coverage gaps; provider abstraction with feature flags mitigates
DataForSEO / SerpApi	External	No Google AIO/AI Mode data; dual-provider adapter mitigates
Stripe + Razorpay	External	Billing for global + India; Phase 1 uses Stripe only
Google Search Console API	External	Optional enrichment; feature degrades gracefully
Trademark clearance for product name	Internal	Rebrand cost if late; clear before public launch

1.4 Glossary

Term	Meaning
Prompt	A natural-language query submitted to an AI engine or search engine on behalf of a brand
Observation	One sampled response from one engine for one prompt at one time, stored raw and parsed
Mention rate	Share of observations in which the brand is named
Recommendation rate	Share of observations in which the brand is presented as a recommended choice
Citation	A URL an engine references in support of its answer
SOV	Share of voice: brand mentions divided by brand plus tracked competitor mentions
Recommendation	An evidence-backed suggested action generated by the platform
Task	A recommendation accepted into the action workflow with owner, due date and status
Experiment	A before/after measurement window around a completed change
Tenant	An Organization; the unit of data isolation

2. Product overview

The platform answers one question: when customers search for a problem, category, product or recommendation, how visible is a brand across Google and AI answer engines, and what exactly should the brand do next.

2.1 Category and positioning

- Category: AI Visibility & Brand Search Intelligence (combines AI search visibility, Brand SERP optimization and GEO/AEO).
- One-line definition: a platform that shows exactly how visible a brand is across Google and AI answer engines, and turns that into an evidence-backed, measurable action plan.

- Ideal customer: mid-market B2B SaaS and D2C brands, and the agencies serving them; enterprise tier from Phase 3.
- Differentiator: citation-opportunity discovery, root-cause diagnosis and closed-loop experiments on reproducible, uncertainty-aware measurement, plus an e-commerce product-title wedge and India-language depth.

2.2 Core product loop

```
flowchart LR
  D[Discover<br/>prompts] --> M[Monitor<br/>engines]
  M --> A[Analyze<br/>SOV, competitors]
  A --> G[Diagnose<br/>root cause]
  G --> R[Recommend<br/>evidence-backed]
  R --> E[Execute<br/>tasks]
  E --> X[Experiment<br/>before/after]
  X --> ME[Measure<br/>re-check]
  ME --> L[Learn<br/>what worked]
  L --> D
```

Every screen answers four questions: what happened, why, what should I do, did my action work.

2.3 Goals (Phase 1, first 90 days)

Goal	Target	Measured by
Monitoring live across 4 AI engines + Google surfaces	Day 30	Observations stored daily for 100% of active prompts
Visibility metrics with confidence intervals	Day 45	Dashboard renders mention, recommendation, citation rate, SOV with CI and n
Recommendation engine and task workflow	Day 75	100% of recommendations carry evidence links; tasks re-checked after completion
10 paying design-partner brands (India D2C + agencies)	Day 90	Active subscriptions
Cost per monitored prompt per engine per day	⩽ $0.02 blended	FinOps dashboard

2.4 Non-goals

- Full SEO suite parity with Semrush or Ahrefs (backlinks index, keyword database at scale).
- Consumer-UI scraping of ChatGPT, Gemini or Perplexity.
- Auto-publishing content to customer CMSs without human approval (Phase 3 pilot only, approval-gated).
- Paid media, social scheduling, or review-response automation.

2.5 Success metrics

Metric	Phase 1 target	Phase 3 target
Active brands	25	400
Prompts monitored per day	5,000	250,000
Recommendation acceptance rate	30%	50%
Tasks with measured impact	60% of completed	90% of completed
Gross margin	55%	72%
Net revenue retention	n/a	110%

3. Users, roles and permissions

Access is scoped Organization → Workspace → Brand, with six roles enforced by RBAC in the API layer and by row-level security in Postgres.

3.1 Hierarchy

```
flowchart TD
  O[Organization<br/>tenant, billing] --> W1[Workspace<br/>agency client or
BU]
  O --> W2[Workspace]
  W1 --> B1[Brand]
  W1 --> B2[Brand]
  B1 --> D[Domains, products,<br/>competitors, locales]
  B1 --> P[Prompts, engines,<br/>schedules]
```

An Organization is the tenant and billing unit. A Workspace groups brands (an agency client, a business unit, a region). Users belong to an Organization and receive roles per Workspace.

3.2 Roles

Role	Scope	Intended user
Org Owner	Organization	Founder, agency principal; billing, SSO, org settings
Org Admin	Organization	Ops lead; all workspaces, user management, no billing
Workspace Admin	Workspace	Account manager; brands, members, integrations in that workspace
Editor	Workspace or Brand	SEO/content/PR practitioner; prompts, tasks, experiments
Viewer	Workspace or Brand	CMO, client stakeholder; read-only dashboards and reports
API Service Account	Workspace	Machine access with scoped API key

3.3 Permission matrix

Capability	Owner	Org Admin	WS Admin	Editor	Viewer	Service
Manage billing and plan	Yes	No	No	No	No	No
Configure SSO, SCIM, data residency	Yes	Yes	No	No	No	No
Create or delete workspace	Yes	Yes	No	No	No	No
Invite users, assign roles	Yes	Yes	WS only	No	No	No
Create or archive brand	Yes	Yes	Yes	No	No	Scoped
Edit brand entities, competitors, locales	Yes	Yes	Yes	Yes	No	Scoped
Create, edit, pause prompts	Yes	Yes	Yes	Yes	No	Scoped
Trigger on-demand audit	Yes	Yes	Yes	Yes	No	Scoped
View dashboards, reports	Yes	Yes	Yes	Yes	Yes	Read
Accept recommendation into task, assign, close	Yes	Yes	Yes	Yes	No	Scoped
Create experiment	Yes	Yes	Yes	Yes	No	Scoped
Configure alerts	Yes	Yes	Yes	Yes	Own only	No
White-label and client portal settings	Yes	Yes	Yes	No	No	No

Capability	Owner	Org Admin	WS Admin	Editor	Viewer	Service
Export raw observations	Yes	Yes	Yes	No	No	Read
View audit log	Yes	Yes	WS only	No	No	No
Manage API keys	Yes	Yes	WS only	No	No	No

3.4 Rules

- Every API request resolves to (user or service account, organization, workspace, brand set) and sets `app.tenant_id` on the database session; RLS policies filter every tenant-scoped table.
- Roles are additive across workspaces; the highest role wins within a workspace.
- Client portal users (Phase 2) are Viewers restricted to one workspace with white-label branding.
- All role changes, exports and deletions are written to the immutable audit log.

4. Functional requirements

Sixteen feature modules make up the product; each lists its user stories, requirements and acceptance criteria. Priority: P0 = Phase 1 launch blocker, P1 = Phase 1 or early Phase 2, P2 = Phase 2, P3 = Phase 3.

F01 Onboarding and brand setup (P0)

User story. As a marketing lead, I add my brand, domain, products and competitors in under 10 minutes and see a first visibility snapshot within an hour.

Requirements:

- Sign up with email + password or Google; email verification; org created on first login.
- Brand wizard: name, aliases (min 1, max 20), primary domain, additional domains, category, products (name + URL), competitors (name + domain, max 10 on Starter, 25 on Pro, unlimited Enterprise), target countries and languages.
- Auto-suggest competitors from a seed engine query; user confirms each.
- Auto-generate 50 starter prompts (F02) on brand creation and queue an initial monitoring run.
- Optional: connect Google Search Console (OAuth) and a Shopify or product feed for the e-commerce wedge (F16).

Acceptance criteria:

- Brand creation with all required fields completes in a single wizard; validation errors are inline.
- Initial run for 50 prompts across enabled engines completes within 60 minutes of creation.
- Alias matching correctly distinguishes `Nike` from `NikeKids` in the benchmark set (precision ≥ 0.97).

F02 Prompt intelligence (P0)

User story. As a content director, I want the platform to discover the questions customers ask about my category so I monitor what matters, not what I guessed.

Requirements:

- Generate prompts across 10 intent classes: informational, commercial, transactional, comparison, alternative, recommendation, problem-based, industry-specific, persona-specific, geographic.
- Sources: LLM generation seeded by brand/category/products; Google autocomplete and People Also Ask via SERP provider; Search Console queries (if connected); optional CSV import; manual entry.
- Deduplicate with embeddings (cosine ≥ 0.92 = duplicate); cluster by intent, topic, funnel stage, persona, product, geography, language.
- Localized prompts per country/language including Hindi and Hinglish variants.
- Prompt states: suggested, active, paused, archived; plan-based active limits (Starter 100, Pro 500, Agency 2,000 per workspace, Enterprise custom).
- Prompt volume estimate (low/medium/high) from keyword APIs where available, labelled as estimate.

Acceptance criteria:

- 50 starter prompts generated in ≤ 90 s with ≤ 5% near-duplicates on the benchmark set.
- Each prompt shows intent, cluster, locale and source badge.
- Plan limits enforced at API and UI.

F03 Multi-engine monitoring pipeline (P0)

User story. As an SEO director, I need daily observations for every active prompt on every enabled engine, with full raw data for audit.

Requirements:

- Engines Phase 1: ChatGPT (OpenAI Responses API with `web_search`), Gemini (grounding), Perplexity (Sonar), Claude (web search tool), Google organic + AI Overviews + AI Mode (SERP provider). Phase 2: Copilot, Grok, Google Images, YouTube, news.
- Cadence per plan: daily (Pro/Agency/Enterprise), every 3 days (Starter); on-demand run limited by quota.
- Samples per prompt per engine: 3 (Starter, Pro), 7 (Agency, Enterprise), configurable up to 10.
- Store for each observation: prompt, engine, model, model version, timestamp, locale, device hint, request parameters, full raw response, citations, parsed extraction, pipeline version.
- Provider adapters with retries, exponential backoff, circuit breakers, rate-limit budgets per provider, and a fallback SERP provider.
- Idempotent job execution; a re-run never creates duplicate observations for the same (prompt, engine, scheduled slot, sample index).

Acceptance criteria:

- 99% of scheduled observations complete within the 6-hour daily window.
- Raw payload retrievable for any observation within 12 months.
- Provider outage for 1 engine does not block other engines (verified by chaos test).

F04 Answer analysis and extraction (P0)

User story. As an analyst, I want each AI answer parsed into brand presence, position, recommendation status, competitors, sentiment, context and citations, with the reason the brand was absent when it was.

Requirements:

- Deterministic pass first: alias and domain matching (exact, fuzzy with edit distance ⩽ 1 on tokens ⩾ 5 chars, URL-domain match), citation URL parsing and domain normalization.
- LLM extraction pass (small model, JSON schema): mention position (ordinal in list or paragraph), recommendation status (recommended, listed, mentioned, negative), competitors named, sentiment (positive, neutral, negative, mixed) with confidence, context summary (⩽ 40 words), absence reason class (not retrieved, competitor dominance, category mismatch, off-topic answer).
- Human-labelled benchmark of 1,000 observations; mention precision/recall ⩾ 0.95, recommendation F1 ⩾ 0.85, sentiment agreement ⩾ 0.80.
- Re-extraction on pipeline version bump without re-querying engines.

Acceptance criteria:

- Every observation has an extraction record with `pipeline_version` and `model_used`.
- Benchmark thresholds enforced in CI; a regression blocks deploy.

F05 Visibility metrics and share of voice (P0)

User story. As a CMO, I want honest visibility numbers I can compare week over week.

Metric definitions (all per brand, engine, prompt set, period):

Metric	Formula
Mention rate	observations with brand mention / eligible observations
Recommendation rate	observations with status recommended / eligible observations
Average position	mean ordinal position when mentioned
Citation rate	observations citing a brand-owned domain / eligible observations
Share of voice	brand mentions / (brand mentions + Σ tracked competitor mentions)
Prompt coverage	prompts with ≥ 1 mention in the period / active prompts
Sentiment index	(positive − negative) / mentioned observations
Brand Visibility Score	weighted composite of the above plus Google visibility, entity consistency, SERP ownership; weights visible and versioned

Requirements:

- Every metric displays n and a 95% Wilson confidence interval; deltas that do not clear the interval are shown as "no significant change".
- Composite score labelled "proprietary diagnostic, not an official engine metric" wherever shown.
- Aggregations pre-computed nightly into a metrics mart; on-demand recompute for custom filters within 5 s at p95.
- Metric methodology page with formulas and version history.

Acceptance criteria:

- Numbers on dashboard, report and API agree exactly for the same filter.
- Changing sample count changes CI width, not the point estimate method.

F06 Citation intelligence and opportunity engine (P1)

User story. As a PR manager, I want to know which sites AI engines cite for my category and which of them mention competitors but not us.

Requirements:

- Citation table: domain, URL, page title, engine, frequency, first/last seen, cited for which prompts, which brands were mentioned in the citing answer, source type (vendor, review site, publication, community, docs, other), authority estimate (third-party metric where licensed, labelled estimate).
- Gap detection: sources citing ≥ 2 tracked competitors and 0 brand citations.
- Opportunity records: source, why relevant (prompts and clusters), competitors present, recommended action (outreach, listing, content contribution, correction), suggested content brief, priority score, status.
- Crawl cited pages (respecting robots.txt) to extract on-page brand mentions and structured data.

Acceptance criteria:

- For a brand with 5 competitors and 300 prompts, opportunity list is generated within 24 h of the first full run.
- Each opportunity links to ≥ 1 raw observation as evidence.

F07 Competitor intelligence (P1)

Requirements:

- Per competitor: mention rate, recommendation rate, SOV, citation domains, sentiment, prompt coverage, week-over-week deltas.
- Visibility change feed: competitor gained or lost ≥ 10 percentage points on ≥ 10 prompts within 14 days, with linked evidence (new cited pages, new citing domains).
- Weekly crawl of competitor domains for new pages (sitemap diff) and new structured data.

Acceptance criteria: each change entry lists the prompts, engines and evidence URLs; false-positive rate on the benchmark ≤ 10%.

F08 Entity intelligence and consistency audit (P1)

Requirements:

- Brand entity graph: brand, company, products, people, categories, features, partners, social profiles, review profiles, knowledge sources; editable by Editors; auto-seeded from the brand's site and schema.org markup.
- Consistency audit across website, About page, LinkedIn, Crunchbase, G2, Capterra, Wikipedia/Wikidata (where present), directories: compare name, description, category, founders, location, logo, social links; flag conflicts with severity.
- Schema.org checks: Organization (`name`, `url`, `logo`, `sameAs`, identifiers), WebSite, Product, Article, BreadcrumbList; explain each finding as an optimization signal, never a guaranteed outcome.

Acceptance criteria: audit runs on demand and monthly; each finding has location, observed value, expected value, and fix guidance.

F09 Brand SERP audit (P1)

Requirements: for `[brand]` and `[brand] reviews` in each target locale, classify first-page results (official, social, review, news, competitor, negative, video, image, Knowledge Panel, sitelinks, AI Overview), compute SERP ownership share, and produce a Brand SERP Health Report with findings.

Acceptance criteria: report generated weekly; classification accuracy ≥ 0.9 on benchmark.

F10 Site and AI crawlability audit (P1)

Requirements:

- Crawl up to 500 pages (Starter), 5,000 (Pro), 50,000 (Enterprise) with Playwright rendering for JS sites; respect robots.txt.
- Checks: robots directives for Googlebot, Bingbot, GPTBot, OAI-SearchBot, ClaudeBot, Claude-SearchBot, PerplexityBot, Google-Extended; sitemap validity; canonicals; indexability; titles, metas, H1; structured data validity; author attribution; `llms.txt` presence (labelled unofficial); CDN bot blocking detection via fetch with each bot UA.
- Each finding: severity, affected URLs, why it matters for AI/search discoverability, fix.

Acceptance criteria: crawl of 5,000 pages completes in ≤ 2 h; findings deduplicated by rule and URL pattern.

F11 Content gap and optimization engine (P1)

Requirements:

- Content gaps: clusters where competitors are mentioned or cited and the brand is absent or weak; output target query, intent, recommended page type, evidence, competitors, citation sources, suggested structure, internal links, priority.
- URL optimizer: fetch page, score topic coverage, intent fit, entity coverage, evidence, author attribution, structure, FAQs, tables, direct answers; output add, remove, rewrite, evidence-to-add lists.
- AI Citability Score (0–100), proprietary diagnostic, components visible, weights versioned.

Acceptance criteria: URL analysis completes in ⩽ 60 s p95; every suggestion cites the observation or competitor page that motivates it.

F12 Recommendation engine and action plans (P0)

User story. As a marketing lead, I open the app and see the three most valuable things to do today, each with evidence and a plan.

Requirements:

- Recommendation types: create content, optimize page, fix entity inconsistency, fix crawlability, pursue citation source, respond to reputation issue, expand prompts, add competitor.
- Each recommendation: problem, evidence (linked observations, metrics, pages), root cause class, action steps, expected outcome (qualitative plus which metric should move), difficulty, priority score, suggested owner role.
- Generation: deterministic rules produce candidates from diagnostics; a large model drafts the plan text; the generator/critic loop (ported from the ADK demo) reviews for evidence completeness and forbidden claims (no guarantees, no manipulation).
- Priority score = expected reach (prompts affected × volume estimate) × competitor pressure × feasibility, weights visible.
- Recommendations regenerate weekly; accepted ones become Tasks (F13); dismissed ones record a reason.

Acceptance criteria: 100% of recommendations pass the critic check for evidence links and forbidden language before display; top-3 panel loads in ⩽ 1.5 s.

F13 Action management (P0)

Requirements:

- Task fields: title, linked recommendation, owner, due date, status (open, in progress, done, verified, impact measured), notes, attachments (URLs).

- Status flow with automatic re-check: on done, schedule re-crawl of affected URLs and a 14-day measurement window on affected prompts; on window close, compute before/after deltas with CI and set impact label (improved, no significant change, declined).
- Board and list views; assignment notifications by email; Slack/Teams webhooks (Phase 2).

Acceptance criteria: every task marked done gets an impact record within 16 days; impact visible on the task and in Learn analytics.

F14 Experiments (P2)

Requirements: hypothesis, treatment prompts and control prompts, change description, before window (≥ 14 days), after window (≥ 14 days), metrics (mention, recommendation, citation, SOV) with CI; result labelled association not causation; minimum sample enforced (≥ 7 samples/prompt/day for experiments).

Acceptance criteria: an experiment cannot be marked significant unless the after-window CI excludes the before-window point estimate.

F15 Alerts, reporting and agency mode (P1)

Requirements:

- Alerts on rolling windows only: mention rate drop ≥ 15 pts over 7 days, competitor displacement on ≥ 10 prompts, new negative mention cluster, new competitor citation source with authority above threshold, Brand SERP change; channels email, Slack, webhook; per-user preferences; digest mode.
- Reports: CMO summary, SEO, content, PR, agency client report; scheduled weekly/monthly; PDF and shareable link; white-label logo, colours, custom sender, custom domain (Phase 2); every report prints methodology version, n and CI.
- Agency mode: workspaces per client, client portal with Viewer access, usage limits per workspace, consolidated billing.

Acceptance criteria: no alert fires on a single observation; PDF report of 20 pages renders in ≤ 30 s; white-label report contains no platform branding.

F16 E-commerce product visibility wedge (P0)

User story. As a catalog manager, I connect my product feed and see which products are invisible for the generic terms shoppers use, with rewritten titles.

Requirements:

- Ingest products via Shopify OAuth, Google Merchant feed URL, or CSV (title, description, attributes, URL, price, category).
- Per product: infer generic shopper vocabulary (ported from the ADK keyword agent, temperature raised to 0.4 with 3 samples and vote), rank keywords generic-first, drop brand terms.
- Compare against top-ranking titles from SERP provider (Google Shopping and organic) for the rank-1 keyword; optional marketplace check via licensed provider (Phase 2), never unlicensed scraping.
- Output per product: missing keywords, suggested title (⩽ 150 chars, brand kept, no keyword stuffing), suggested description bullet points; bulk export CSV; push to Shopify as draft on approval (Phase 2).
- Batch: up to 1,000 products per run; progress and partial results.

Acceptance criteria: 1,000-product run completes in ⩽ 45 min; suggested titles pass a rule check (no duplicate tokens, brand preserved, length) 100% of the time.

F17 Reputation monitoring (P2)

Requirements: ingest reviews (G2, Capterra, Trustpilot, Google Business via licensed APIs), Reddit and forums (official APIs), news (news API), YouTube (Data API); classify sentiment, topic, reach; suggested response drafts labelled as drafts; no posting from the platform.

F18 Admin, billing and settings (P0)

Requirements: plans Starter, Pro, Agency, Enterprise; Stripe (global) and Razorpay (India, Phase 2); usage metering on prompts, samples, crawls, exports; overage warnings at 80% and 100%; invoices; org and workspace settings; API keys; audit log viewer; data export and deletion requests (DPDP/GDPR).

5. User flows

Seven primary flows cover the product loop; each is a state the UI, API and job pipeline must support end to end.

5.1 Onboarding to first insight

```
flowchart LR
  S[Sign up] --> V[Verify email] --> O[Create org] --> B[Brand wizard]
  B --> C[Confirm competitors] --> P[Review 50 prompts]
  P --> R[Initial run queued] --> D[Dashboard fills<br/>within 60 min]
```

While the initial run executes, the dashboard shows progress per engine and a sample of completed observations rather than an empty state.

5.2 Daily monitoring (system flow)

```
sequenceDiagram
  participant Sch as Cloud Scheduler
  participant Orch as Orchestrator job
  participant Q as Pub/Sub
  participant W as Engine workers
  participant X as Extraction workers
  participant DB as Postgres / BigQuery
  Sch->>Orch: 02:00 tenant-local trigger
  Orch->>Q: enqueue (prompt, engine, sample) messages
  Q->>W: pull with rate budget per provider
  W->>DB: write raw observation to GCS + row
  W->>Q: enqueue extraction
  Q->>X: extract, classify, cite
  X->>DB: write parsed record, update metrics mart
  X-->>Orch: window complete -> alerts + recommendations
```

5.3 Diagnose to recommendation

1. User opens a losing cluster from the dashboard ("Where am I losing").
2. Cluster view shows mention rate per engine, competitor SOV, citation domains, and absence reasons with counts.
3. Diagnosis panel lists root causes ranked by evidence: missing content, not cited by key sources, entity inconsistency, crawler blocked.
4. Each root cause links to its recommendation; user accepts, dismisses with reason, or snoozes.

5.4 Recommendation to measured impact

```
stateDiagram-v2
  [*] --> Recommended
  Recommended --> Accepted: accept
  Recommended --> Dismissed: dismiss + reason
  Accepted --> InProgress: assign owner
  InProgress --> Done: mark done
  Done --> Verified: re-crawl confirms change
  Verified --> ImpactMeasured: 14-day window closes
  ImpactMeasured --> [*]
```

The Learn view aggregates impact outcomes by recommendation type so the priority weights improve with evidence.

5.5 Experiment

1. Editor creates an experiment from a task or from scratch: hypothesis, treatment prompts, control prompts, change description.
2. System enforces ⩾ 14-day before window already captured at ⩾ 7 samples/prompt/day; otherwise the experiment waits.
3. Editor marks the change live; after window runs 14–28 days.
4. Result page: before vs after per metric with CI, control comparison, and the label significant, not significant, or inconclusive.

5.6 Agency client reporting

1. Workspace Admin sets white-label branding once per workspace.
2. Schedules a monthly CMO report per brand; recipients include client Viewers.
3. Report renders as PDF and shareable link; client portal shows the same data read-only.

5.7 E-commerce product run

1. Catalog manager connects Shopify or uploads CSV.
2. Selects up to 1,000 products and target locale.
3. Run shows progress; results table lists product, generic keyword, competing titles, missing keywords, suggested title, and a rule-check badge.
4. Manager exports CSV or (Phase 2) approves push to Shopify drafts.

6. UI/UX requirements

The interface is a Next.js 15 application (App Router, React 19, TypeScript) with a token-based design system; every screen answers what happened, why, what to do, and did it work.

6.1 Design principles

- Premium, quiet, data-first: one accent colour, neutral surfaces, no decorative animation; transitions ⩽ 150 ms.
- Uncertainty is visible: every rate shows n and CI; deltas below significance render grey with the label "no significant change".
- Evidence is one click away: every number opens the observations behind it.

- Plain language; no jargon without a hover definition; proprietary scores carry a persistent "diagnostic" tag.
- AI-generated text is labelled and kept short; the user sees evidence tables before prose.

6.2 Information architecture

Nav item	Purpose	Key screens
Overview	Today's picture and actions	Dashboard, Critical issues, Top actions
Visibility	Where the brand appears	Engines, Prompts, Clusters, Google surfaces, Brand SERP
Competitors	Comparison and changes	Leaderboard, Change feed, Competitor detail
Citations	Sources and opportunities	Citation table, Opportunity list, Source detail
Site	Technical and content health	Crawl findings, Entity graph, Consistency audit, URL optimizer
Actions	Recommendations, tasks, experiments, learn	Recommendation inbox, Task board, Experiments, Learn
Products	E-commerce wedge	Catalog runs, Product results
Reports	Scheduled and ad-hoc reports	Report builder, Schedules, Archive
Settings	Org, workspace, brand, billing, API, alerts	Settings pages

A global workspace and brand switcher sits in the top bar; agencies see workspaces grouped by client.

6.3 Dashboard specification

- Row 1: six metric tiles (Brand Visibility Score, AI Mention Rate, AI Recommendation Rate, Citation Rate, Google Visibility, Competitive Share) each with value, 7-day and 30-day delta, sparkline, n and CI on hover.
- Row 2: Winning and losing clusters (top 5 each) with engine chips and SOV bars.
- Row 3: Competitor movement (last 14 days) and Critical issues / Opportunities / Improvements counters.
- Row 4: Top 3 recommended actions with evidence summary and one-click accept.

- Filters: engine, locale, date range, prompt cluster; filter state in URL.

6.4 Screen inventory

Screen	Priority	Notes
Auth (sign up, sign in, SSO, reset)	P0	Google OAuth, magic link
Brand wizard (5 steps)	P0	Resumable
Dashboard	P0	Spec above
Prompt manager (table, cluster tree, bulk actions)	P0	Virtualized table, 10k rows
Prompt detail (observations over time per engine, raw answer viewer)	P0	Diff view between samples
Cluster detail with diagnosis panel	P1	
Competitor leaderboard and detail	P1	
Citation table and opportunity list	P1	
Crawl findings and URL optimizer	P1	
Entity graph editor	P1	Node graph, inline edit
Recommendation inbox, Task board, Task detail	P0	Kanban and list
Experiment setup and results	P2	
Report builder and archive	P1	
Product run and results	P0	
Settings suite (12 pages)	P0	
Client portal (white-label)	P2	Separate route group

6.5 Components and states

- Tables: server-side pagination, sortable, column chooser, saved views, CSV export; loading skeletons; empty states with a next action; error states with retry and request id.
- Charts: line, bar, stacked bar, distribution; consistent colour per engine; accessible data table toggle.
- Every long operation (runs, crawls, reports) shows progress with per-stage counts and estimated completion.

6.6 Accessibility and responsiveness

- WCAG 2.2 AA: contrast ⩾ 4.5:1, keyboard navigation for all actions, focus order, ARIA on custom controls, reduced-motion support.
- Breakpoints 1440, 1280, 1024, 768; dashboard and reports fully usable at 768; tablet-first for client portal; mobile web read-only for dashboard, tasks and alerts.
- Internationalization: UI strings externalized (en-IN, en-US at launch; hi-IN Phase 2); numbers, dates and currency localized.

6.7 Performance budgets

- First contentful paint ⩽ 1.5 s on 4G; dashboard interactive ⩽ 2.5 s; route transitions ⩽ 300 ms; JS bundle per route ⩽ 250 kB gzipped.

7. Data model

Transactional data lives in Cloud SQL for PostgreSQL 16 with pgvector; observations and metrics are mirrored to BigQuery for analytics; raw payloads live in Cloud Storage. Every tenant-scoped table carries `org_id` and is protected by row-level security.

7.1 Entity relationships

```
flowchart TD
  Org[organization] --> WS[workspace] --> Br[brand]
  Br --> Dom[domain]
  Br --> Prod[product]
  Br --> Comp[competitor]
  Br --> Loc[locale]
  Br --> Pr[prompt] --> Cl[prompt_cluster]
  Pr --> Obs[observation] --> Ext[extraction]
  Ext --> Men[mention]
  Ext --> Cit[citation] --> Src[source] --> Pg[page]
  Br --> Rec[recommendation] --> Task[task] --> Imp[impact]
  Br --> Exp[experiment]
```

Audit findings, alerts, reports and users hang off brand, workspace and organization respectively (table below).

7.2 Tables and key fields

Table	Key fields	Notes
organization	id, name, plan, region (asia-south1 or us-central1), billing_customer_id, settings jsonb, created_at	Tenant root
workspace	id, org_id, name, branding jsonb, usage_limits jsonb	Agency client or BU
user	id, email, auth_provider, status, mfa_enabled	Global; membership via user_role
user_role	user_id, org_id, workspace_id nullable, brand_id nullable, role	Composite unique
api_key	id, org_id, workspace_id, hashed_key, scopes, last_used_at, expires_at	Hash only
brand	id, org_id, workspace_id, name, aliases text[], category, primary_domain_id, status	
domain	id, org_id, brand_id, host, is_primary, crawl_config jsonb	
product	id, org_id, brand_id, name, url, sku, attributes jsonb, source	Feed or manual
competitor	id, org_id, brand_id, name, aliases text[], domains text[]	
locale	id, org_id, brand_id, country, language, city nullable	
prompt	id, org_id, brand_id, locale_id, text, intent, funnel_stage, persona, source, status, volume_est, embedding vector(768), cluster_id	HNSW index on embedding
prompt_cluster	id, org_id, brand_id, label, topic, intent	

Table	Key fields	Notes
engine	id, code (chatgpt, gemini, perplexity, claude, copilot, google_organic, google_aio, google_aimode), provider, default_model	Reference
schedule	id, org_id, brand_id, engine_id, cadence, samples, next_run_at	
observation	id, org_id, brand_id, prompt_id, engine_id, model, model_version, locale_id, sample_index, scheduled_slot, requested_at, params jsonb, raw_uri (GCS), status, cost_micros	Unique (prompt_id, engine_id, scheduled_slot, sample_index); partitioned by month
extraction	id, observation_id, pipeline_version, model_used, brand_mentioned bool, position int, rec_status, sentiment, sentiment_conf, context, absence_reason, parsed jsonb	One current per observation
mention	id, org_id, observation_id, entity_type (brand or competitor), entity_id, position, rec_status, sentiment	Indexed (org_id, entity_id, observation_id)
citation	id, org_id, observation_id, url, domain, position, source_id	
source	id, org_id, domain, source_type, authority_est, first_seen, last_seen	Per org to allow annotations
page	id, org_id, source_id nullable, domain_id nullable, url, title, fetched_at, content_uri, schema jsonb, citability_score	
serp_result	id, org_id, brand_id, locale_id, query, fetched_at, position, url, result_type, features jsonb	

Table	Key fields	Notes
audit_finding	id, org_id, brand_id, audit_type, rule_code, severity, urls text[], observed, expected, guidance, status	
entity_node / entity_edge	graph of brand entities	jsonb properties
metric_daily	org_id, brand_id, engine_id, locale_id, cluster_id nullable, entity_id, date, n, mentions, recommendations, citations, position_sum, sentiment_pos, sentiment_neg, sov	Mart; also in BigQuery
recommendation	id, org_id, brand_id, type, title, problem, evidence jsonb, root_cause, plan jsonb, expected_outcome, difficulty, priority_score, status, generated_at, model_used, critic_passed	
task	id, org_id, brand_id, recommendation_id, owner_user_id, due_date, status, notes	
impact	id, task_id, window_start, window_end, before jsonb, after jsonb, label	
experiment	id, org_id, brand_id, hypothesis, treatment_prompt_ids, control_prompt_ids, change_at, before_window, after_window, result jsonb, status	
alert_rule / alert_event	rule config; fired events with evidence and delivery status	
report / report_run	definition; rendered PDF uri, recipients, status	

Table	Key fields	Notes
usage_ledger	org_id, workspace_id, meter (prompt_obs, crawl_page, export, report), quantity, period	Billing
audit_log	id, org_id, actor_id, action, target_type, target_id, before, after, ip, at	Append-only

7.3 Indexing and partitioning

- All tenant tables: B-tree on (org_id, id); observation and extraction partitioned monthly by requested_at with indexes on (org_id, brand_id, prompt_id, engine_id, requested_at desc).
- prompt.embedding: HNSW (m=16, ef_construction=200) with cosine distance; queries always filtered by org_id first.
- metric_daily: composite primary key covers the dashboard filter order (org_id, brand_id, date, engine_id, locale_id, entity_id).
- citation: index on (org_id, domain, observation_id) for citation tables; source unique on (org_id, domain).

7.4 Retention

Data	Hot (Postgres)	Warm (BigQuery)	Cold (GCS)
Raw responses	n/a	n/a	12 months Standard, then Coldline to 36 months (Enterprise configurable)
Observations and extractions	13 months	36 months	Export archive
metric_daily	36 months	60 months	
Audit log	24 months	84 months	
Deleted tenant	purged within 30 days across all tiers; deletion certificate emitted		

8. API specification

One versioned REST API (`/v1`) serves both the web app and external customers; it is FastAPI (Python 3.12) with OpenAPI 3.1 as the contract and generated TypeScript client for the frontend.

8.1 Conventions

- Base URL `https://api.<domain>/v1` ; JSON only; UTC ISO-8601 timestamps; cursor pagination (`limit` , `cursor` , `next_cursor`); field filtering via `fields` ; sorting via `sort=-created_at` .
- Auth: web app uses short-lived JWT (15 min) from Firebase Authentication / Identity Platform with refresh; external clients use `Authorization: Bearer <api_key>` with scoped keys, or OAuth 2.0 client credentials (Phase 2).
- Every request carries `X-Request-Id` (echoed) and resolves tenant context; cross-tenant ids return 404, never 403.
- Idempotency: `Idempotency-Key` header required on POST that create runs, tasks, experiments, exports; keys stored 24 h.
- Errors: RFC 9457 problem+json with `type` , `title` , `status` , `detail` , `request_id` , `errors[]` for validation.
- Rate limits per API key: Starter 60 rpm, Pro 300 rpm, Agency 600 rpm, Enterprise custom; `429` with `Retry-After` ; headers `X-RateLimit-Limit` , `X-RateLimit-Remaining` .
- Versioning: path version; breaking changes only in a new major; deprecations announced 6 months ahead with `Sunset` header.

8.2 Resources

Resource	Methods	Notes
/orgs, /orgs/{id}	GET, PATCH	Owner/Admin
/workspaces	GET, POST, PATCH, DELETE	
/users, /invitations, /roles	GET, POST, PATCH, DELETE	
/brands, /brands/{id}	GET, POST, PATCH, DELETE	Archive instead of hard delete
/brands/{id}/competitors, /domains, /products, /locales	CRUD	

Resource	Methods	Notes
/brands/{id}/prompts	GET, POST, PATCH, bulk POST	Bulk import up to 5,000
/brands/{id}/prompts/suggest	POST	Returns suggested prompts, async job
/brands/{id}/clusters	GET	
/brands/{id}/runs	GET, POST	On-demand run; returns job id
/observations	GET, GET /{id}, GET /{id}/raw	Signed URL for raw
/brands/{id}/metrics	GET	Filters engine, locale, cluster, date; returns n and CI
/brands/{id}/sov	GET	Brand and competitors
/brands/{id}/citations, /sources, /opportunities	GET, PATCH (status)	
/brands/{id}/competitor-changes	GET	
/brands/{id}/audits	GET, POST	site, entity, serp, crawlability
/brands/{id}/entity-graph	GET, PUT	
/pages/analyze	POST	URL optimizer, async
/brands/{id}/recommendations	GET, POST /{id}/accept, /dismiss	
/tasks	CRUD, POST /{id}/transition	Status transitions validated
/experiments	CRUD, POST /{id}/start, /close	
/alerts/rules, /alerts/events	CRUD, GET	
/reports, /reports/{id}/runs	CRUD, GET	PDF signed URL
/products/runs	GET, POST	E-commerce wedge
/exports	POST, GET	Async; signed URL, 24 h expiry

Resource	Methods	Notes
/webhooks	CRUD	Subscriptions
/usage	GET	Metered usage per period
/audit-log	GET	Admin

8.3 Jobs and webhooks

- Long operations return `202` with `{job_id, status_url}`; `/jobs/{id}` reports queued, running, succeeded, failed with progress counts.
- Webhook events: `run.completed`, `observation.batch_ready`, `recommendation.created`, `task.impact_measured`, `alert.fired`, `report.rendered`, `export.ready`, `audit.completed`. Payload signed with HMAC-SHA256 (`X-Signature-256`); retries with exponential backoff up to 24 h; dead-letter visible in settings.

8.4 Documentation

- Public docs generated from OpenAPI with examples in curl, Python, TypeScript; a Postman collection; changelog; sandbox tenant with seeded data.

9. External integrations

All third-party access goes through a provider adapter layer with a common interface (`query`, `cost`, `health`, `rate_budget`) so any provider can be swapped or disabled by feature flag without code changes elsewhere.

Integration	Purpose	Method	Phase	Failure mode
OpenAI Responses API (`web_search`)	ChatGPT observations	Official API, per-request cost logged	1	Engine paused, others continue
Google Gemini API grounding	Gemini observations	Official API via Vertex AI	1	Same
Perplexity Sonar	Perplexity observations	Official API	1	Same

Integration	Purpose	Method	Phase	Failure mode
Anthropic Messages API (web search tool)	Claude observations	Official API	1	Same
Microsoft Copilot	Copilot observations	Licensed provider or official API when available; otherwise not offered	2	Not offered
DataForSEO (primary), SerpApi (fallback)	Google organic, AIO, AI Mode, autocomplete, PAA, Shopping	Licensed SERP APIs	1	Automatic failover
Google Search Console API	Query and page performance enrichment	OAuth per brand	1	Feature hidden when disconnected
Shopify Admin API	Product feed and draft push	OAuth app	1 (read), 2 (write)	Manual CSV
Google Merchant feed	Product ingest	Feed URL fetch	1	CSV
Review and community APIs (G2, Capterra, Trustpilot, Reddit, YouTube Data, news API)	Reputation	Official/licensed	2	Source hidden
Playwright crawler (own)	Site and cited-page crawling	Cloud Run jobs, robots.txt respected	1	Partial findings
Stripe	Global billing	Billing portal, webhooks	1	Read-only billing UI
Razorpay	India billing (INR, UPI)	Webhooks	2	Stripe fallback
SendGrid or Resend	Transactional email	API	1	Retry queue

Integration	Purpose	Method	Phase	Failure mode
Slack, Microsoft Teams	Alerts and task notifications	Incoming webhooks, OAuth app	2	Email fallback
Identity Platform (Firebase Auth), SAML/OIDC	Authentication, SSO	Managed	1 (email, Google), 3 (SAML, SCIM)	

Legal and data-handling rules: no consumer-UI scraping of AI engines; SERP data only from licensed providers; crawler identifies itself with a named user agent and contact URL, honours robots.txt and 429s; all provider keys in Secret Manager, never in tenant scope.

10. AI/ML architecture

LLMs are used only where rules cannot do the job; every LLM call is routed by task to the cheapest model that meets the benchmark, emits structured JSON, and is logged with model, version, tokens and cost.

10.1 Task routing

Task	Method	Default model	Fallback
Alias and domain matching	Deterministic (normalized tokens, edit distance, URL parse)	none	none
Citation URL extraction	Deterministic from provider payload	none	none
Mention position, recommendation status, sentiment, context, absence reason	LLM, JSON schema	Gemini 2.5 Flash-Lite	Claude Haiku
Prompt generation	LLM with brand context	Gemini 2.5 Flash	GPT-5 mini
Prompt dedupe and clustering	Embeddings + HDBSCAN	text-embedding (Vertex) 768-d	

Task	Method	Default model	Fallback
Cluster labelling	LLM on cluster centroid samples	Flash-Lite	
Source type classification	Rules (domain lists) then LLM for unknown	Flash-Lite	
Page citability scoring	Rules (structure, schema, author, references) + LLM for claim clarity	Flash	
Recommendation drafting	LLM large + generator/critic loop	Gemini 2.5 Pro or Claude Sonnet	
Product keyword inference (wedge)	LLM, 3 samples, majority vote	Flash	
Report narrative	LLM with metric tables only	Flash	

10.2 Pipeline stages

```
flowchart LR
  Q[Query engine] --> R[Store raw]
  R --> D[Deterministic pass]
  D --> L[LLM extraction]
  L --> V[Validate schema<br/>+ sanity rules]
  V --> M[Metrics mart]
  M --> Dg[Diagnostics rules]
  Dg --> Rc[Recommendation draft]
  Rc --> Cr[Critic check]
  Cr --> Out[Publish]
```

10.3 Sampling and variability

- Default samples per prompt per engine per day: 3 (Starter, Pro), 7 (Agency, Enterprise); experiments require 7.
- Requests use provider defaults for temperature (consumer-like), not 0, so samples reflect real variance; parameters stored per observation.
- Metrics are computed on observations, never on one "representative" answer; CI uses the Wilson interval on the sample count.

10.4 Evaluation and drift

- Golden benchmark: 1,000 labelled observations across engines and locales, refreshed quarterly; CI gate on mention precision/recall ≥ 0.95, recommendation F1 ≥ 0.85, sentiment agreement ≥ 0.80.
- Model change protocol: new model runs shadow extraction on 5% of traffic for 7 days; promote only if benchmark and shadow agreement pass; `pipeline_version` bumps and historical re-extraction offered.
- Weekly drift report: engine answer length, citation counts, refusal rates per engine; anomalies alert the data team.

10.5 Cost controls

- Prompt caching for shared system prompts; batch APIs for extraction backlog (50% discount where offered); Flash-Lite as default extractor; large models only for recommendation text.
- Per-tenant daily budget with soft stop and admin alert; per-provider spend caps in Secret-Manager-backed config.
- Target blended cost per observation ≤ $0.012 (engine call + extraction) at Phase 1 volumes.

11. Non-functional requirements

The platform targets 99.9% availability for the web app and API, 99% on-time completion for daily monitoring, and linear cost scaling to 250,000 observations per day.

Area	Requirement	Verification
Availability	Web and API 99.9% monthly; pipeline 99% of scheduled observations within the 6 h window	SLO dashboards, monthly review
Latency	API reads p95 ≤ 400 ms, writes p95 ≤ 800 ms; dashboard metrics query p95 ≤ 1.5 s; raw observation fetch p95 ≤ 2 s	Load tests at 3× forecast
Throughput	250,000 observations/day (Phase 3) with 2,000 concurrent extraction workers max	Pipeline soak test
Scalability	Stateless services autoscale 0–N; database vertical to 32 vCPU then read replicas; BigQuery for analytics	Capacity plan reviewed quarterly

Area	Requirement	Verification
Reliability	Idempotent jobs, at-least-once delivery with dedupe, circuit breakers per provider, dead-letter queues with replay	Chaos tests each release
Data quality	Every observation has raw payload and extraction with version; benchmark gates in CI; daily completeness report	Automated
Data residency	India tenants in asia-south1, US/EU tenants in us-central1 (EU region europe-west1 Phase 3); no cross-region replication of tenant data	Terraform policy, audit
Compliance	DPDP Act 2023 and Rules 2025; GDPR (DPA, SCCs, data subject requests within 30 days); SOC 2 Type I by month 12	External audit
Accessibility	WCAG 2.2 AA	Automated axe + manual audit per release
Browser support	Last 2 versions of Chrome, Edge, Safari, Firefox	CI matrix
Maintainability	80% unit coverage on domain logic; ADRs for architecture decisions; conventional commits	CI gate
Portability	All infra as Terraform; containers only; no GCP-specific code outside adapters	Review
Capacity headroom	30% above forecast on database and queues	Monitoring

11.1 Capacity forecast

Phase	Brands	Prompts/day	Observations/day	Raw storage/month	Postgres size
1 (month 3)	25	5,000	60,000	60 GB	50 GB

Phase	Brands	Prompts/day	Observations/day	Raw storage/month	Postgres size
2 (month 6)	120	40,000	500,000	500 GB	400 GB
3 (month 12)	400	250,000	3,000,000	3 TB	2.5 TB (partition pruning, BigQuery offload)

Assumes 4 engines, 3 samples average, 20 KB per raw response.

12. Security requirements

Security follows defense in depth: tenant isolation in the database, least-privilege service identities, managed secrets, encrypted transport and storage, and an immutable audit trail.

12.1 Identity and access

- Authentication via Google Cloud Identity Platform: email + password (breached-password check, minimum 12 chars), Google OAuth, magic link; MFA (TOTP) optional for all, required for Org Owner and Admin; SAML/OIDC SSO and SCIM provisioning in Phase 3.
- Sessions: 15-minute access JWT, 30-day rotating refresh token bound to device; revocation list checked on refresh.
- API keys: 256-bit random, shown once, stored as SHA-256 hash, scoped by workspace and capability, expiring, rotatable; usage logged.
- Service-to-service: Workload Identity, no long-lived keys; each service its own service account with minimal IAM roles.

12.2 Tenant isolation

- Postgres row-level security on every tenant table keyed by `org_id`; application role cannot bypass RLS; connection sets `app.org_id` from the verified token per transaction.
- Storage paths prefixed `orgs/{org_id}/`; signed URLs scoped and expiring ⩽ 1 h; BigQuery row-level access policies mirror RLS.
- Enterprise option: dedicated Cloud SQL instance and GCS bucket per tenant in the tenant's region.
- Automated isolation test suite attempts cross-tenant reads for every endpoint on each release.

12.3 Data protection

- TLS 1.2+ everywhere; HSTS; Cloud SQL private IP; encryption at rest with Google-managed keys by default, CMEK (Cloud KMS) for Enterprise.
- PII minimization: the platform stores business data; user PII limited to name, email, IP for audit; raw AI responses may contain third-party names and are treated as confidential tenant data.
- Secrets in Secret Manager with rotation; no secrets in env files, images or logs; log redaction for tokens and emails.
- Data subject requests (access, correction, erasure) fulfilled through settings within 30 days; tenant deletion purges all tiers within 30 days.

12.4 Application security

- OWASP ASVS level 2; input validation with Pydantic and Zod; output encoding; CSRF protection on cookie sessions; CSP, X-Frame-Options, Referrer-Policy headers.
- Prompt-injection defense: content fetched from the web or returned by engines is data, never instructions; extraction prompts are templated with delimiters; recommendation critic rejects any output containing instructions to the platform or unsupported claims.
- SSRF protection in the crawler and URL analyzer: allow-list of schemes, block private ranges and metadata endpoints, DNS re-resolution check.
- Dependency scanning (Dependabot, pip-audit, npm audit), container scanning (Artifact Analysis), SAST (Semgrep) and secret scanning in CI; critical findings block release.
- Annual third-party penetration test from Phase 2; bug bounty policy page.

12.5 Audit and governance

- Append-only audit log for auth events, role changes, exports, deletions, settings changes, API key events; retained 24 months hot, 7 years cold; exportable for Enterprise.
- Cloud Audit Logs enabled for admin and data access on production projects; forwarded to a separate security project.
- Incident response runbook: severity levels, 1-hour acknowledgement for Sev1, customer notification within 72 h for confirmed personal-data breaches (DPDP and GDPR).

12.6 Compliance roadmap

Milestone	Target
Privacy policy, DPA template, cookie consent, data-residency controls	Phase 1 launch
DPDP compliance review, GDPR records of processing	Month 6
SOC 2 Type I	Month 12
SOC 2 Type II, ISO 27001 readiness	Month 18–24

13. GCP infrastructure architecture

The platform runs entirely on serverless and managed GCP services: Cloud Run for web, API and workers; Cloud SQL, Memorystore, BigQuery and Cloud Storage for data; Pub/Sub, Cloud Tasks and Cloud Scheduler for the pipeline; all defined in Terraform.

13.1 Architecture decisions

Decision	Choice	Alternatives considered	Rationale
Compute	Cloud Run (services and jobs)	GKE Autopilot	Scale-to-zero, no cluster ops for a small team; GKE revisited if sustained > 300 concurrent workers
Pipeline orchestration	Cloud Scheduler + Pub/Sub + Cloud Tasks + Cloud Run jobs	Temporal, Airflow, Cloud Workflows	Simple fan-out with per-provider rate control; Temporal adopted in Phase 3 if multi-step sagas grow
Transactional DB	Cloud SQL for PostgreSQL 16 (Enterprise Plus), pgvector	AlloyDB, Spanner	Mature, RLS, pgvector; AlloyDB if read-heavy analytics on Postgres grows
Analytics	BigQuery (existing skill and data)	ClickHouse on GCE	Managed, existing BigQuery use, cheap storage

Decision	Choice	Alternatives considered	Rationale
Raw payload store	Cloud Storage with lifecycle	Storing JSON in Postgres	Cost and size
Cache and rate limits	Memorystore for Redis	Firestore	Token buckets, session cache, idempotency keys
Auth	Identity Platform	Auth0, Clerk	Native GCP, SAML/OIDC, low cost
Frontend hosting	Cloud Run (Next.js SSR) behind global external HTTPS LB + Cloud CDN	Firebase Hosting, Vercel	Single vendor, private egress to API
Rendering crawler	Playwright on Cloud Run jobs (2 vCPU, 4 GB)	Browserless SaaS	Cost control, data locality
Secrets	Secret Manager	Vault	Managed, IAM-integrated
IaC	Terraform with Cloud Build	Pulumi	Team familiarity, module ecosystem

13.2 Project and organization layout

Project	Contents
`bsi-shared`	Artifact Registry, Cloud Build, Terraform state bucket, DNS, org-level IAM
`bsi-security`	Aggregated logs, SCC, audit sinks, KMS key rings
`bsi-dev`	Full stack, small instances, synthetic data
`bsi-staging`	Production-like, anonymized subset, blue/green rehearsals
`bsi-prod-in`	Production for India-resident tenants, asia-south1
`bsi-prod-us`	Production for US/EU tenants, us-central1 (europe-west1 added Phase 3)

Folders: `env/dev`, `env/staging`, `env/prod`; organization policies enforce domain-restricted sharing, uniform bucket-level access, disabled service-account key creation, and required VPC Service Controls on prod data services.

13.3 Runtime topology

```
flowchart TD
  U[Users] --> LB[Global HTTPS LB<br/>Cloud Armor + CDN]
  LB --> WEB[Cloud Run: web<br/>Next.js SSR]
  LB --> API[Cloud Run: api<br/>FastAPI]
  API --> SQL[(Cloud SQL Postgres<br/>pgvector, HA)]
  API --> RED[(Memorystore Redis)]
  API --> PS[Pub/Sub topics]
  SCH[Cloud Scheduler] --> ORC[Cloud Run job: orchestrator]
  ORC --> PS
  PS --> ENG[Cloud Run: engine workers]
  PS --> EXT[Cloud Run: extraction workers]
  PS --> CRL[Cloud Run job: crawler]
  ENG --> GCS[(Cloud Storage raw)]
  EXT --> SQL
  EXT --> BQ[(BigQuery marts)]
  ENG -. NAT egress .-> EXTAPI[Provider APIs]
```

Each production region is a copy of this topology inside its own project and VPC; there is no cross-region tenant data flow.

13.4 Networking

- One VPC per environment with subnets for Cloud Run (Direct VPC egress), Cloud SQL private services access, and Redis; no public IPs on data services.
- Cloud NAT with reserved static IPs for provider allow-listing; separate NAT for the crawler with its own IP pool.
- Global external Application Load Balancer with managed certificates, HTTP/2 and HTTP/3, Cloud CDN for static assets and public report links, Cloud Armor with OWASP preconfigured rules, rate limiting per IP and per API key header, and geo rules if required by a tenant.
- Internal traffic between web and API over serverless VPC with IAM-authenticated invocations; provider egress only from worker services.
- VPC Service Controls perimeter around Cloud SQL, GCS, BigQuery and Secret Manager in production.
- Cloud DNS for `app.`, `api.`, `reports.` and customer white-label CNAMEs (certificate map for custom domains, Phase 2).

13.5 Compute sizing (production, Phase 1 → Phase 3)

Service	CPU / memory	Min–max instances	Concurrency	Notes
web	1 vCPU / 1 GB	1–20	80	CPU always allocated for SSR
api	2 vCPU / 2 GB	2–50	40	Startup CPU boost
engine-worker	1 vCPU / 1 GB	0–400	8	Pull subscription, provider budgets in Redis
extraction-worker	1 vCPU / 2 GB	0–600	10	Batch mode nightly
crawler (job)	2 vCPU / 4 GB	0–50 tasks	1	Playwright
orchestrator (job)	1 vCPU / 1 GB	scheduled	1	Per tenant-region at 02:00 local
report-renderer (job)	2 vCPU / 4 GB	0–20	1	Headless Chrome PDF

13.6 Data services

- Cloud SQL Enterprise Plus, PostgreSQL 16, 4 vCPU/16 GB at Phase 1 scaling to 16 vCPU/64 GB, regional HA, 1 read replica from Phase 2, automated backups, PITR 7 days, connection pooling via Cloud SQL Auth Proxy sidecar and PgBouncer.
- Memorystore for Redis Standard tier 5 GB with replica.
- BigQuery datasets `observations`, `metrics`, `usage` per region; partitioned by date, clustered by org_id and brand_id; scheduled queries for the nightly mart.
- Cloud Storage buckets: `raw-{region}` (Standard → Coldline at 12 months), `reports-{region}`, `exports-{region}` (24 h lifecycle), `crawl-{region}`; uniform access, versioning on raw.
- Pub/Sub topics: `monitor.requests`, `monitor.results`, `extract.requests`, `crawl.requests`, `alerts.evaluate`, each with a dead-letter topic and 7-day retention.

13.7 Edge, CDN and static assets

- Next.js static assets and public report pages served through Cloud CDN with immutable cache headers; HTML responses `no-store`.
- Image and PDF downloads via signed URLs; CDN caches public share links keyed by token.

13.8 Cost posture

- Serverless scale-to-zero for workers; committed-use discount on Cloud SQL from Phase 2; BigQuery on-demand with a 1 TB/day project quota; Coldline lifecycle on raw payloads; budget alerts at 50/80/100% per project.

14. Environments

Three environments run the same Terraform modules with different sizing and data policies; code promotes dev → staging → production by tag, never by rebuild.

Attribute	Development	Staging	Production
GCP project	bsi-dev	bsi-staging	bsi-prod-in, bsi-prod-us
Purpose	Feature work, integration tests	Release rehearsal, performance and security tests, customer demos	Customer traffic
Data	Synthetic tenants, seeded benchmark observations	Anonymized copy of 5% of production, refreshed weekly	Real
Provider APIs	Sandbox keys, daily spend cap $20	Live keys, spend cap $200/day, reduced sampling	Live
Cloud SQL	2 vCPU, zonal	4 vCPU, regional HA	4–16 vCPU, regional HA, replica
Scale limits	Workers max 20	Workers max 100	Full
Deploy trigger	Merge to `main`	Tag `rc-*`	Tag `v*` + manual approval
Access	All engineers	Engineers, QA, PM	Ops on-call, break-glass with approval

Attribute	Development	Staging	Production
Feature flags	All on	Production config	Production config
Retention	30 days	90 days	Per policy

Local development uses Docker Compose (Postgres with pgvector, Redis, Pub/Sub emulator, GCS emulator) with recorded provider fixtures so tests need no API keys.

15. CI/CD and DevOps

A monorepo with trunk-based development, Cloud Build pipelines, Terraform for all infrastructure, and canary releases on Cloud Run give a release cadence of daily to dev, twice weekly to staging, and weekly to production.

15.1 Repository layout

Path	Contents
apps/web	Next.js frontend
apps/api	FastAPI service
apps/workers	Engine, extraction, crawler, report, orchestrator
packages/core	Domain models, metric formulas, extraction schemas (shared Python)
packages/adk-lib	Ported generator/critic loop and keyword agent from the ADK demo
packages/ui	Design system components
infra/terraform	Modules and environment roots
eval	Benchmark datasets and evaluation harness (extends the ADK evalset)

15.2 Pipeline

```
flowchart LR
  PR[Pull request] --> CI[Lint, type check,<br/>unit, contract tests]
  CI --> SEC[SAST, dependency,<br/>secret scan]
  SEC --> BLD[Build images<br/>Artifact Registry]
  BLD --> DEV[Deploy dev<br/>on merge]
  DEV --> INT[Integration + AI<br/>benchmark gate]
  INT --> STG[Deploy staging<br/>on rc tag]
  STG --> E2E[E2E, load,<br/>isolation tests]
  E2E --> PRD[Canary 10% prod<br/>on v tag + approval]
  PRD --> FULL[100% after<br/>30 min SLO check]
```

15.3 Practices

- Images built once, signed with Binary Authorization, and promoted by digest; SBOM attached.
- Database migrations via Alembic, expand-and-contract only, run as a Cloud Run job before traffic shift; every migration has a tested down path.
- Terraform plan on PR, apply on merge through Cloud Build with state in GCS and locking; drift detection nightly.
- Feature flags (config in Postgres, cached in Redis) for every user-facing feature and provider; kill switches per engine.
- Rollback: Cloud Run traffic shift to previous revision within 2 minutes; migrations designed to stay compatible one version back.
- Release notes generated from conventional commits; changelog published for API consumers.
- On-call rotation with runbooks per alert; postmortems within 5 business days for Sev1 and Sev2.

16. Observability

Google Cloud Observability (Logging, Monitoring, Trace, Error Reporting) with OpenTelemetry instrumentation gives one trace per request or job across web, API and workers, and SLO-based alerting routed through PagerDuty.

16.1 Logging

- Structured JSON logs with `request_id`, `org_id`, `brand_id`, `job_id`, `trace_id`; severity levels; PII and secrets redacted at the SDK.

- Retention: 30 days in Logging, 400 days in a BigQuery log sink for audit and analytics; security sinks to the security project.
- Log-based metrics for provider errors, rate-limit hits, extraction validation failures, RLS denials.

16.2 Metrics and SLOs

SLI	SLO	Alert (burn rate)
API availability (non-5xx)	99.9% / 30 days	2% budget in 1 h, 5% in 6 h
API latency p95 reads	≤ 400 ms	14.4× over 5 min
Daily monitoring completion within window	99% of scheduled observations	Window at 80% elapsed with < 90% done
Extraction validation pass rate	≥ 99%	< 97% over 1 h
Provider error rate per engine	≤ 2%	> 5% over 15 min
Cost per observation	≤ target per plan	Daily > 120% of target
Alert delivery latency	≤ 5 min	> 15 min
Report render success	99%	Any failure spike

Custom metrics: observations per engine per status, extraction latency, tokens and cost per model, queue depth and age, crawler pages per minute, benchmark scores per pipeline version.

16.3 Tracing and errors

- OpenTelemetry auto-instrumentation for FastAPI, SQLAlchemy, Redis, Pub/Sub, HTTP clients; 10% sampling in production, 100% for errors; Cloud Trace with exemplars linked from latency charts.
- Error Reporting groups exceptions; Sentry optional for frontend session replay (PII-masked).
- Synthetic checks every minute from 3 regions on sign-in, dashboard load, and API health; uptime page.

16.4 Dashboards

- Service health (per service, per region), Pipeline (queues, completion, provider health), AI quality (benchmark, validation, drift), FinOps (spend by provider, model, tenant), Security (auth failures, RLS denials, WAF blocks), Business (active brands, prompts, recommendations accepted, impact outcomes).

17. Backup, disaster recovery and business continuity

Recovery objectives: RPO 15 minutes and RTO 4 hours for a regional incident; RPO 24 hours and RTO 24 hours for loss of an entire GCP region.

Asset	Backup	Location	Restore test
Cloud SQL	Automated daily backups, PITR 7 days, weekly export to GCS retained 90 days	Same region plus cross-region copy of exports (India → asia-south2 Delhi; US → us-east1)	Monthly restore drill in staging
Cloud Storage raw and reports	Object versioning, dual-region bucket for raw in Phase 2	Dual-region within jurisdiction	Quarterly sample restore
BigQuery	Time travel 7 days, weekly snapshot tables retained 90 days	Same region	Quarterly
Redis	Not source of truth; rebuilt from Postgres		
Secrets and config	Secret Manager versions, Terraform state versioned	Multi-region	With infra drill
Infrastructure	Terraform reproducible in a new project within 2 h		Semi-annual full rebuild drill

- Regional failover: Cloud SQL regional HA handles zonal loss automatically; for full-region loss, restore the latest export into the paired region within the same jurisdiction and repoint DNS; pipeline resumes from Pub/Sub backlog and idempotent schedules.
- Data-residency constraint: India tenants never fail over outside India; US/EU tenants never into India.

- Business continuity: provider outage playbooks (pause engine, notify customers, extend window), key-person coverage on runbooks, quarterly game days.

18. Testing and QA strategy

Quality is enforced by automated gates at every pipeline stage plus a standing AI benchmark; nothing reaches production without passing the isolation, benchmark and E2E suites.

18.1 Test layers

Layer	Scope	Tooling	Gate
Unit	Domain logic, metric formulas, matchers, schema validation	pytest, Vitest	80% coverage on packages/core; 100% on metric formulas
Contract	OpenAPI conformance, generated client	Schemathesis, Prism	PR
Integration	API + Postgres + Redis + Pub/Sub emulator; provider adapters against recorded fixtures (VCR)	pytest, testcontainers	Merge to main
AI benchmark	1,000 labelled observations; prompt generation dedupe; clustering stability; recommendation critic	eval harness (extends ADK evalset)	Dev → staging
Tenant isolation	Automated cross-tenant access attempts on every endpoint and storage path	Custom suite	Staging → prod
E2E	Onboarding, monitoring run, dashboard, recommendation to task, report, product run	Playwright	Staging
Load	3× forecast API traffic; 250k observation soak	k6, custom	Monthly and before major releases

Layer	Scope	Tooling	Gate
Chaos	Provider outage, queue lag, DB failover, region loss drill	Scripts, game days	Quarterly
Security	SAST, DAST (ZAP), dependency and container scans, annual pentest	CI, external	Release
Accessibility	axe automated + manual keyboard and screen-reader pass	CI, QA	Release
Visual regression	Key screens	Playwright snapshots	PR

18.2 AI-specific test cases

- Mention detection: exact, alias, possessive, plural, substring collision (Nike vs NikeKids), transliteration (Hindi script brand names), competitor name inside URL only.
- Recommendation status: listed vs recommended vs "not recommended"; negation handling.
- Citation extraction: provider-specific payload shapes, redirect and tracking-parameter normalization, duplicate domains.
- Sentiment: mixed answers, comparative statements, sarcasm flagged as low confidence.
- Clustering: same input yields the same clusters across runs (Jaccard ≥ 0.9); new prompts join existing clusters without relabelling.
- Variability: metric CI widths match expected values for known sample sizes; alerts do not fire on single-observation swings (property-based tests).
- Forbidden-claims critic: fixtures containing "guarantee ranking", "buy reviews", "fake traffic" must be rejected 100%.
- Model change: shadow run agreement ≥ 0.95 on mention, ≥ 0.85 on recommendation before promotion.
- Duplicate and idempotency: re-running a slot creates zero new observations; replayed webhooks deduplicate.
- Rate limits and failures: 429, 5xx, timeouts, malformed payloads per provider; circuit breaker opens and closes correctly.

18.3 Release criteria

- All gates green; benchmark scores at or above the previous release; zero open Sev1/Sev2 bugs; isolation suite 100%; performance budgets met on staging; rollback rehearsed for any migration; release notes and runbook updates merged.

19. Cost model and FinOps

All figures are approximate estimates from public list prices as of September 2026 and must be re-validated before pricing decisions; third-party API spend dominates and is metered per tenant.

19.1 Unit cost per observation (estimate)

Component	Approximate cost per observation	Basis
ChatGPT via Responses API with web search	$0.012	~$10 per 1,000 search calls plus ~1.5k tokens
Gemini grounding	$0.015	$14 per 1,000 grounded queries after free tier, plus tokens
Perplexity Sonar	$0.007	~$5 per 1,000 searches plus tokens
Claude with web search	$0.013	$10 per 1,000 searches plus tokens
Google organic + AIO via DataForSEO	$0.002	~$0.6–$1.2 per 1,000
Extraction (Flash-Lite, ~2k tokens)	$0.0006	List price
Storage, queue, compute, logging	$0.0005	Amortized
Blended (4 AI engines + Google, per prompt-sample)	~$0.010 per engine-sample	

19.2 Monthly cost per brand by prompt count (estimate)

Assumes 4 AI engines + Google, 3 samples, daily cadence, 30 days.

Prompts	Engine-samples/month	API + extraction	Infra share	Total per brand
100	45,000	~$450	~$15	~$465
500	225,000	~$2,250	~$40	~$2,290
1,000	450,000	~$4,500	~$70	~$4,570
10,000	4,500,000	~$45,000	~$500	~$45,500

At 7 samples multiply API cost by 2.3; at every-3-day cadence divide by 3. These figures show daily 3-sample monitoring on 5 surfaces is expensive at scale, which drives the packaging rules below.

19.3 Packaging and cost controls

- Starter: 100 prompts, 2 engines, every 3 days, 3 samples → ~$30/month API cost; price ~$99.
- Pro: 300 prompts, 4 engines, daily, 3 samples → ~$1,080/month API cost at list; target ⩽ $400 through weekly cadence on low-volume clusters, batch extraction, caching and provider negotiation; price ~$499–$799.
- Agency and Enterprise priced per prompt-engine-day with volume tiers and committed provider contracts.
- Levers: adaptive cadence (daily only for high-volume and high-volatility clusters, weekly otherwise), sample count by plan, provider volume discounts, batch APIs, Flash-Lite extraction, dedupe of identical answers across samples before extraction.
- FinOps: per-tenant usage ledger and daily spend dashboard; margin per tenant reviewed monthly; hard caps per provider per day; alerts at 80% of budget.

19.4 Platform infrastructure (GCP, monthly estimate)

Phase	Cloud Run	Cloud SQL	Redis	BigQuery + GCS	LB, CDN, NAT, Armor	Logging, Monitoring	Total
1	$300	$450	$120	$80	$150	$100	~$1,200
2	$1,200	$1,400	$250	$400	$300	$300	~$3,850
3	$4,500	$3,800	$500	$1,800	$600	$800	~$12,000

Two production regions in Phase 3 roughly split the totals above rather than doubling them; committed-use discounts on Cloud SQL and Cloud Run reduce Phase 3 by ~20%.

20. Delivery plan

Three phases take the product from foundations to enterprise readiness in 12 months; Phase 1 ships a paid MVP in 90 days.

20.1 Phases and milestones

Phase	Window	Scope	Exit criteria
1a Foundations	Days 1–30	Monorepo, Terraform for dev and staging, auth, org/workspace/brand model with RLS, brand wizard, prompt intelligence v1, monitoring pipeline for ChatGPT, Gemini, Perplexity, Claude and Google (DataForSEO), raw storage, extraction v1, metrics mart	Initial run for a new brand completes in 60 min; benchmark gates live in CI
1b Insight	Days 31–60	Dashboard, prompt and cluster views, SOV and CI, citation intelligence v1, competitor leaderboard and change feed, crawlability and entity audits, product wedge (CSV and Shopify read)	10 design partners onboarded; numbers reconcile across UI, API, reports
1c Action	Days 61–90	Recommendation engine with critic, task workflow and impact windows, alerts on rolling windows, CMO/SEO/content reports, Stripe billing, production in asia-south1 and us-central1, SLOs and on-call	Paid launch; 99% pipeline completion for 14 consecutive days
2 Scale	Months 4–6	Experiments, agency workspaces and white-label, client portal, Razorpay, Slack/Teams, Copilot and Grok, reputation monitoring v1, Search Console enrichment, Hindi/Hinglish prompts, public API v1 and webhooks, Shopify draft push, DPDP/GDPR review	120 brands; NRR tracking live; pentest complete

Phase	Window	Scope	Exit criteria
3 Enterprise	Months 7–12	SSO/SAML/SCIM, CMEK, dedicated instances, europe-west1, approval workflows, advanced attribution, autonomous optimization pilot (approval-gated), SOC 2 Type I	400 brands; SOC 2 Type I report

20.2 Team and cadence

- Pod of 9 (PM, designer, 2 backend, 2 frontend, data/AI, DevOps, QA); two-week sprints; weekly release to production from Phase 1c.
- Design partners give weekly feedback; a monthly steering review checks goals in section 2.3 and the cost targets in section 19.

20.3 Risks

Risk	Likelihood	Impact	Mitigation
Provider pricing or terms change (search tools, grounding)	High	Cost model breaks	Adapter layer, dual providers, adaptive cadence, contractual caps in customer plans
API answers differ from consumer UI answers	High	Trust	Label method per observation, publish methodology, validate quarterly against a small licensed UI panel
Third-party SERP provider legal exposure	Medium	Google surface data loss	Two licensed providers; degrade to Search Console and organic only
Extraction quality drift after model updates	Medium	Wrong metrics	Shadow runs, benchmark gates, pipeline versioning with re-extraction
Cost per prompt exceeds plan price	Medium	Margin	Usage metering, hard caps, cadence and sample levers, Flash-Lite default
Tenant data leak	Low	Severe	RLS, isolation suite, VPC-SC, pentest

Risk	Likelihood	Impact	Mitigation
Team capacity for 90-day scope	Medium	Slip	Scope cut order: reputation, experiments, Copilot, white-label before core loop
Name and trademark	Medium	Rebrand	Clearance before public launch

20.4 Open questions

- [ ] Final product name and domain after trademark search
- [ ] Launch pricing for Starter, Pro and Agency after design-partner interviews
- [ ] Whether Copilot can be observed through an official or licensed channel by Phase 2; otherwise it is not offered
- [ ] Which review and community APIs are licensed for India-focused brands (Google Business, Amazon.in reviews)
- [ ] EU region timing based on demand from EU design partners
- [ ] Whether the e-commerce wedge should support Amazon.in marketplace ranking through a licensed provider in Phase 2

20.5 Reuse from the existing ADK application

Existing asset	New home	Change
Generator/critic comparison loop	packages/adk-lib, used by recommendation critic and product wedge	Prompts rewritten for evidence and forbidden-claim checks
keyword_finding_agent logic	Product wedge keyword inference	Temperature 0.4, 3 samples with vote; non-English support added
google_search grounding agent	Gemini engine adapter and product competitor lookup	Wrapped behind the provider interface
BigQuery connector	Analytics mart loaders	Parameterised queries kept; catalog table replaced by product entity
Eval harness and evalset discipline	eval/ benchmark suite	Extended from trajectory match to labelled extraction benchmarks

Existing asset	New home	Change
Selenium site-browsing agent	Not carried forward	Replaced by licensed providers and the Playwright crawler for owned and cited pages